

CMPUT 267: Machine Learning I · Fall 2026

Midterm 1 Review

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Midterm 1

Date

Tuesday, September 29

Setting

In class

Writing Time

70 minutes

Course Weight

27.5%

The exam has 20 questions. Allow a few minutes before and after the writing time to distribute and collect the exam packages.

Question Format

Questions use a multiple-choice format similar to the assignment questions.

- 1 A question may have one correct answer or several correct answers.
- 2 The question will not tell you how many choices are correct.
- 3 Select every correct choice to receive full credit.

Materials and Answer Sheet

Provided

Exam booklet with the formula sheet and answer sheet attached

Bring

A pencil and eraser

(Non programmable) calculators are allowed.



Only answers written on the attached answer sheet are marked. Write all chosen letters for multiple-choice questions and the requested values for numerical questions. Write clearly and erase changes completely.

Example Answer Sheet

CMPUT 267 Fall 2026
Midterm 1 Preparation Exam
Answer Sheet

Last Name: First Name:

For multiple-choice questions, write **all chosen letters** (A, B, C, D). For numerical questions, write the requested value(s). Exact fractions such as a/b are accepted; decimal conversion is not required.

Q.	Answer	Q.	Answer
1	_____	11	_____
2	_____	12	_____
3		13	
4	_____	14	_____
5	_____	15	_____
6		16	
7	_____	17	_____
8	_____	18	_____
9	$\mathbb{E}[f(X)] =$ _____	19	$w_1 =$ _____ $w_2 =$ _____
10	$\hat{L}(f) =$ _____	20	$\mathbb{E}[N] =$ _____

Formula Sheet: Page 1

Formula Sheet		
Integration		
$\int_a^b x^d dx = \frac{x^{d+1}}{d+1} \Big _a^b = \frac{b^{d+1} - a^{d+1}}{d+1} \quad \text{for } d \neq -1$		
Derivatives		
$f(x) = x^a,$	$f'(x) = \frac{df}{dx}(x) = ax^{a-1}$	
$f(x) = \exp(x),$	$f'(x) = \frac{df}{dx}(x) = \exp(x)$	
$f(x) = \log(x),$	$f'(x) = \frac{df}{dx}(x) = \frac{1}{x}$	
$f(x) = g(h(x)), \quad u = h(x)$	$f'(x) = \frac{df}{dx}(x) = \frac{dg}{du} \frac{du}{dx}(x) = g'(u)h'(x) \quad \triangleright \text{Chain rule}$	
$f(x) = g(x)h(x),$	$f'(x) = \frac{df}{dx}(x) = g'(x)h(x) + g(x)h'(x) \quad \triangleright \text{Product rule}$	
Probability		
Univariate:	$\mathbb{P}(X \in \mathcal{E})$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{E}} p(x) & \text{if } X \text{ is discrete} \\ \int_{\mathcal{E}} p(x) dx & \text{if } X \text{ is continuous} \end{cases}$
Multivariate:	$\mathbb{P}(X \in \mathcal{E}_X, Y \in \mathcal{E}_Y)$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{E}_X} \sum_{y \in \mathcal{E}_Y} p(x, y) & \text{if } X, Y \text{ are discrete} \\ \int_{\mathcal{E}_X} \int_{\mathcal{E}_Y} p(x, y) dy dx & \text{if } X, Y \text{ are continuous} \\ \int_{\mathcal{E}_X} \sum_{y \in \mathcal{E}_Y} p_{Y X}(y x) p_X(x) dx & \text{if } X \text{ is continuous, } Y \text{ is discrete} \\ \sum_{x \in \mathcal{E}_X} \int_{\mathcal{E}_Y} p_{Y X}(y x) p_X(x) dy & \text{if } X \text{ is discrete, } Y \text{ is continuous} \end{cases}$
Marginal pmf:	$p_X(x)$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{y \in \mathcal{E}_Y} p(x, y) & \text{if } Y \text{ is discrete} \\ \int_Y p(x, y) dy & \text{if } Y \text{ is continuous} \end{cases}$
Marginal:	$\mathbb{P}_X(X \in \mathcal{E}_X)$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{E}_X} p_X(x) & \text{if } X \text{ is discrete} \\ \int_{\mathcal{E}_X} p_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$
Conditional pmf:	$p_{Y X}(y x)$	$\stackrel{\text{def}}{=} \frac{p(x, y)}{p_X(x)} \quad \text{such that } p_X(x) > 0$
Conditional:	$\mathbb{P}_{Y X}(Y \in \mathcal{E}_Y X = x)$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{y \in \mathcal{E}_Y} p_{Y X}(y x) & \text{if } Y \text{ is discrete} \\ \int_{\mathcal{E}_Y} p_{Y X}(y x) dy & \text{if } Y \text{ is continuous} \end{cases}$
Product Rule:	$p(x, y)$	$= p_{Y X}(y x) p_X(x)$
Bayes' Rule:	$p_X(y x y)$	$= \frac{p_{Y X}(y x) p_X(x)}{p_Y(y)}$
Independence:	$p(x_1, \dots, x_n)$	$= p_{X_1}(x_1) \cdots p_{X_n}(x_n)$
1		

Formula Sheet: Page 2

Distribution	Parameters	pmf or pdf	Expectation and Variance
Bernoulli	$\alpha \in [0, 1]$	$p(x) = \alpha^x(1 - \alpha)^{1-x}, x \in \{0, 1\}$	$\mathbb{E}[X] = \alpha, \text{Var}[X] = \alpha(1 - \alpha)$
Discrete Uniform	$n \in \mathbb{N}$	$p(x) = \frac{1}{n}, x \in \{1, \dots, n\}$	$\mathbb{E}[X] = \frac{n+1}{2}, \text{Var}[X] = \frac{n^2-1}{12}$
Continuous Uniform	$a, b \in \mathbb{R}, a < b$	$p(x) = \frac{1}{b-a}, x \in [a, b]$	$\mathbb{E}[X] = \frac{a+b}{2}, \text{Var}[X] = \frac{(b-a)^2}{12}$
Normal ($\mathcal{N}$)	$\mu \in \mathbb{R}, \sigma^2 > 0$	$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), x \in \mathbb{R}$	$\mathbb{E}[X] = \mu, \text{Var}[X] = \sigma^2$
Laplace	$\mu \in \mathbb{R}, b > 0$	$p(x) = \frac{1}{2b} \exp\left(-\frac{ x-\mu }{b}\right), x \in \mathbb{R}$	$\mathbb{E}[X] = \mu, \text{Var}[X] = 2b^2$

Expectation and Variance

Univariate:	$\mathbb{E}[X]$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{X}} xp(x) & \text{if } X \text{ is discrete} \\ \int_{\mathcal{X}} xp(x)dx & \text{if } X \text{ is continuous} \end{cases}$
Function:	$\mathbb{E}[f(X)]$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{X}} f(x)p(x) & \text{if } X \text{ is discrete} \\ \int_{\mathcal{X}} f(x)p(x)dx & \text{if } X \text{ is continuous} \end{cases}$
Variance:	$\text{Var}[X]$	$\stackrel{\text{def}}{=} \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
Multivariate:	$\mathbb{E}[f(X, Y)]$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} f(x, y)p(x, y) & \text{if } X \text{ and } Y \text{ are discrete} \\ \int_{\mathcal{X}} \int_{\mathcal{Y}} f(x, y)p(x, y)dydx & \text{if } X \text{ and } Y \text{ are continuous} \\ \int_{\mathcal{X}} \sum_{y \in \mathcal{Y}} f(x, y)p_{Y X}(y x)p_X(x)dx & \text{if } X \text{ is continuous, } Y \text{ is discrete} \\ \sum_{x \in \mathcal{X}} \int_{\mathcal{Y}} f(x, y)p_{Y X}(y x)p_X(x)dy & \text{if } X \text{ is discrete, } Y \text{ is continuous} \end{cases}$
Conditional:	$\mathbb{E}[f(Y) X = x]$	$\stackrel{\text{def}}{=} \begin{cases} \sum_{y \in \mathcal{Y}} f(y)p_{Y X}(y x) & \text{if } Y \text{ is discrete} \\ \int_{\mathcal{Y}} f(y)p_{Y X}(y x)dy & \text{if } Y \text{ is continuous} \end{cases}$

Expectation and Variance Properties

- $\mathbb{E}[cX] = c\mathbb{E}[X]$
- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$
- $\text{Var}[c] = 0$
- $\text{Var}[cX] = c^2\text{Var}[X]$

If X and Y are independent:

- $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$

Estimation

Sample Mean:	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
If X_i are i.i.d.:	$\mathbb{E}[\bar{X}] = \mathbb{E}[X_1], \quad \text{Var}[\bar{X}] = \frac{\text{Var}[X_1]}{n}$

Review Topics

Math Notation

Sets, functions, vectors,
sums, and derivatives

Probability

Distributions,
conditioning,
expectation, and
variance

Supervised Learning

Predictors, learners, loss,
risk, and ERM

Optimization

Minimizers, convexity,
and linear regression

 We only go over key points. Read the **course notes and assignments**.

Sets, Tuples, and Functions

Sets

- $x \in \mathcal{X}$: x is an element of $\mathcal{X}$
- $A \subseteq \mathcal{X}$: every element of A is in $\mathcal{X}$
- $\{x \in \mathbb{R} : x \geq 0\}$: set-builder notation

Tuples and products

$$\mathcal{X} \times \mathcal{Y} = \{(x, y) : x \in \mathcal{X}, y \in \mathcal{Y}\}.$$

Functions

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

states that f takes an input from $\mathcal{X}$ and returns an output in $\mathcal{Y}$.

$$\mathcal{A} : (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F}$$

The learner $\mathcal{A}$ takes a dataset of n feature-label pairs and returns a predictor in $\mathcal{F}$.

Function Classes Use Set-Builder Notation

A function class is a set whose elements are functions. For example,

$$\mathcal{F}_d = \left\{ f \mid f(x) = \sum_{j=0}^d w_j x^j, \quad w_0, \dots, w_d \in \mathbb{R} \right\}$$

is the set of polynomial functions of degree at most d .

Reading a Learner Type

Consider

$$\mathcal{A} : (\mathbb{R} \times \mathbb{R})^2 \rightarrow \{f \mid f : \mathbb{R} \rightarrow \mathbb{R}\}.$$

Input

$$D = ((1, 2), (3, 4))$$

Two ordered feature-label pairs form a valid input. The single pair $(1, 2)$ does not.

Output

$$\mathcal{A}(D) = \hat{f}$$

The output must be a function $\hat{f} : \mathbb{R} \rightarrow \mathbb{R}$, not merely a number.

A Valid Learner Need Not Be Useful

The type signature specifies valid inputs and outputs. It does not guarantee good predictions.

$$\mathcal{A}(D) = \hat{f}, \quad \hat{f}(x) = 1 \quad \text{for every } x$$

defines a valid learner because it returns a function in the required function class.

Vectors and Dot Products

For $\mathbf{x}, \mathbf{w} \in \mathbb{R}^d$,

$$\mathbf{x}^\top \mathbf{w} = \sum_{j=1}^d x_j w_j.$$

For example,

$$\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}^\top \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix} = 2(4) - 1(5) + 3(2) = 9.$$

A transpose turns a column vector into a row vector.

Summation Notation

The same finite sum can be written by indexing the values or by naming the values being summed:

$$\sum_{x \in \mathcal{X}} g(x) = \sum_{i=1}^n g(x_i) = g(x_1) + \cdots + g(x_n), \quad \mathcal{X} = \{x_1, \dots, x_n\}.$$

For $g(x) = x^2$ and $\mathcal{X} = \{1, 2, 3\}$,

$$\sum_{x \in \mathcal{X}} g(x) = 1^2 + 2^2 + 3^2 = 14.$$

 For an indexed tuple, every position contributes. Repeated values still appear once for each index.

Integration

Use an integral when a variable can take every real value in an interval.

$$\int_a^b x^d dx = \left. \frac{x^{d+1}}{d+1} \right|_a^b = \frac{b^{d+1} - a^{d+1}}{d+1}, \quad d \neq -1.$$

For example,

$$\int_0^2 x^2 dx = \left. \frac{x^3}{3} \right|_0^2 = \frac{8}{3}.$$

One-Variable Derivative

Use one function throughout this example:

$$g(w_1, w_2) = w_1^2 + w_1 w_2 + 2w_2^2.$$

Fix $w_2 = 2$. This gives a one-variable function of w_1 :

$$h(w_1) = g(w_1, 2) = w_1^2 + 2w_1 + 8, \quad h'(w_1) = 2w_1 + 2.$$

 At $w_1 = 1$, $h'(1) = 4$. The derivative measures the slope as w_1 changes while w_2 stays fixed. The formula sheet lists the standard power, exponential, logarithm, chain, and product rules.

Partial Derivatives

For the same function

$$g(w_1, w_2) = w_1^2 + w_1 w_2 + 2w_2^2,$$

take one derivative for each input while holding the other input fixed.

Hold w_2 fixed:

$$\frac{\partial g}{\partial w_1} = 2w_1 + w_2.$$

At $(1, 2)$, this equals 4.

Hold w_1 fixed:

$$\frac{\partial g}{\partial w_2} = w_1 + 4w_2.$$

At $(1, 2)$, this equals 9.

Gradient

The gradient stacks the partial derivatives into one vector:

$$\nabla g(w_1, w_2) = \begin{bmatrix} \frac{\partial g}{\partial w_1} \\ \frac{\partial g}{\partial w_2} \end{bmatrix} = \begin{bmatrix} 2w_1 + w_2 \\ w_1 + 4w_2 \end{bmatrix}.$$

$$\nabla g(1, 2) = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

- i The first component describes change in the w_1 direction. The second describes change in the w_2 direction. Together, they give the direction of steepest increase.

Outcomes, Events, and Probability

Outcome Space

The set $\mathcal{X}$ of all possible outcomes of an experiment.

Event

Any subset $E \subseteq \mathcal{X}$. The whole space $\mathcal{X}$ is also an event.

Probability

A distribution assigns each event E a number $P(E) \in [0, 1]$, with $P(\mathcal{X}) = 1$.

Random Variable

An unknown value produced by the experiment. Its possible values form the outcome space $\mathcal{X}$.

Events Written with Random Variables

Let X be a die-roll random variable with possible values $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$.

One Value

$$P(X = 4)$$

The singleton event $\{4\}$

Several Values

$$P(X \in \{1, 3, 5\})$$

The event that the roll is odd

A Condition

$$P(X > 4)$$

The event $\{5, 6\}$

 Use uppercase $P(\cdot)$ for event probabilities and lowercase $p(\cdot)$ for a PMF or PDF.

Discrete and Continuous Random Variables

Discrete

The set of possible values is finite or countable. A PMF assigns probability mass to individual values.

$$P(X \in A) = \sum_{x \in A} p_X(x)$$

Continuous

Probabilities come from areas under a PDF. A density value is not probability mass at a point.

$$P(X \in A) = \int_A p_X(x) dx$$

Common Discrete Distributions (on Formula Sheet)

Bernoulli

$$X \sim \text{Bernoulli}(\alpha)$$

$$X \in \{0, 1\} \text{ and } p_X(x) = \alpha^x(1 - \alpha)^{1-x}$$

$$\mathbb{E}[X] = \alpha, \text{Var}(X) = \alpha(1 - \alpha)$$

Discrete Uniform

$$X \in \{1, \dots, m\}$$

$$p_X(x) = 1/m \text{ for every allowed value}$$

$$\mathbb{E}[X] = (m + 1)/2, \text{Var}(X) = (m^2 - 1)/12$$

Common Continuous Distributions (on Formula Sheet)

Distribution	PDF on its support	Mean	Variance
Uniform on $[a, b]$	$p_X(x) = \frac{1}{b - a}$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
Gaussian $\mathcal{N}(\mu, \sigma^2)$	$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$	μ	σ^2
Laplace (μ, b)	$p_X(x) = \frac{1}{2b} \exp\left[-\frac{ x - \mu }{b}\right]$	μ	$2b^2$

Joint, Marginal, and Conditional Distributions

For discrete X and Y :

Joint: $p_{X,Y}(x, y),$

Marginal: $p_X(x) = \sum_y p_{X,Y}(x, y),$

Conditional: $p_{Y|X}(y | x) = \frac{p_{X,Y}(x, y)}{p_X(x)} \quad \text{when } p_X(x) > 0.$

 For continuous variables, replace sums with integrals and PMFs with PDFs.

Multivariable Event Probabilities

For events $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$,

$$P(X \in A, Y \in B) = \begin{cases} \sum_{x \in A} \sum_{y \in B} p_{X,Y}(x, y), & X, Y \text{ discrete,} \\ \int_A \int_B p_{X,Y}(x, y) dy dx, & X, Y \text{ continuous,} \\ \int_A \sum_{y \in B} p_{Y|X}(y | x) p_X(x) dx, & X \text{ continuous, } Y \text{ discrete,} \\ \sum_{x \in A} \int_B p_{Y|X}(y | x) p_X(x) dy, & X \text{ discrete, } Y \text{ continuous.} \end{cases}$$

i Sum over each discrete variable and integrate over each continuous variable.

Product Rule and Bayes' Rule

The product rule reconstructs a joint distribution:

$$p_{X,Y}(x, y) = p_{Y|X}(y | x)p_X(x) = p_{X|Y}(x | y)p_Y(y).$$

Rearranging gives Bayes' rule:

$$p_{Y|X}(y | x) = \frac{p_{X|Y}(x | y)p_Y(y)}{p_X(x)}.$$

Independence

Random variables X and Y are independent when their joint distribution factors:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y.$$

Equivalent consequences include

$$p_{Y|X}(y \mid x) = p_Y(y) \quad \text{and} \quad \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y],$$

where the conditional distribution is defined and the expectations exist.

Functions of Random Variables

If X is a random variable and $g : \mathcal{X} \rightarrow \mathcal{Y}$ is fixed, then $g(X)$ is also a random variable.

$$X \in \mathcal{X} \quad \implies \quad g(X) \in \text{range}(g) \subseteq \mathcal{Y}.$$

For a fair die roll, define

$$g(X) = \mathbb{1}\{X \text{ is even}\}.$$

Then $g(X) \in \{0, 1\}$ and $g(X) \sim \text{Bernoulli}(1/2)$.

Expectation and Variance

Expectation is a probability-weighted average.

Discrete

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x)$$

Continuous

$$\mathbb{E}[g(X)] = \int g(x)p_X(x) dx$$

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Multivariable Expectation

Weight $g(x, y)$ by the joint distribution and sum or integrate over every possible pair.

$$\mathbb{E}[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y) p_{X,Y}(x, y), & X, Y \text{ discrete,} \\ \iint g(x, y) p_{X,Y}(x, y) dy dx, & X, Y \text{ continuous,} \\ \int \sum_y g(x, y) p_{Y|X}(y | x) p_X(x) dx, & X \text{ continuous, } Y \text{ discrete.} \end{cases}$$

Conditional Expectation

Conditioning changes the weights used in the average.

Discrete Y

$$\mathbb{E}[g(Y) \mid X = x] = \sum_y g(y) p_{Y|X}(y \mid x)$$

Continuous Y

$$\mathbb{E}[g(Y) \mid X = x] = \int g(y) p_{Y|X}(y \mid x) dy$$

Useful Expectation and Variance Properties

For random variables X, Y and a constant c ,

$$\begin{aligned}\mathbb{E}[cX] &= c \mathbb{E}[X], & \mathbb{E}[X + Y] &= \mathbb{E}[X] + \mathbb{E}[Y], \\ \text{Var}(c) &= 0, & \text{Var}(cX) &= c^2 \text{Var}(X).\end{aligned}$$

If X and Y are independent,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y], \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Supervised Learning Objects

We observe an i.i.d. dataset

$$D = ((\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)), \quad (\mathbf{X}_i, Y_i) \stackrel{\text{iid}}{\sim} P_{\mathbf{X}, Y}.$$

Predictor

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

Maps a feature vector to a predicted label.

Learner

$$D \mapsto \mathcal{A}(D)$$

Uses the dataset to choose a predictor.

Loss

$$\ell(\hat{y}, y)$$

Measures the cost of one prediction.

Regression and Classification

Regression

The labels are numerical

$$\mathcal{Y} \subseteq \mathbb{R}$$

Classification

The labels identify categories.

$$\mathcal{Y} = \{1, \dots, K\}$$

Risk and Empirical Risk

For a fixed predictor f , the risk is its expected loss on a new example:

$$L(f) = \mathbb{E}_{\mathbf{X}, Y}[\ell(f(\mathbf{X}), Y)].$$

Because $P_{\mathbf{X}, Y}$ is unknown, use the observed dataset $\mathcal{D}$:

$$\hat{L}_{\mathcal{D}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i), y_i).$$

① $\hat{L}_{\mathcal{D}}(f)$ is its estimate from the observed sample.

Empirical Risk Minimization

Two steps of ERM:

Step 1: Estimate

Replace the unknown risk $L(f)$ with empirical risk $\hat{L}_{\mathcal{D}}(f)$.

Step 2: Optimize

Search the chosen function class $\mathcal{F}$ for a predictor with minimum empirical risk.

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_{\mathcal{D}}(f).$$

 The function class controls which predictors the learner is allowed to choose.

The Sample Mean Estimator

For i.i.d. random variables $Z_1, \dots, Z_n$ with mean μ and variance σ^2 ,

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \mathbb{E}[\bar{Z}] = \mu, \quad \text{Var}(\bar{Z}) = \frac{\sigma^2}{n}.$$

Correct on Average

The sample mean has the same expectation as one observation.

More Stable with More Data

Its variance decreases in proportion to $1/n$.

i Empirical risk is a sample mean of the observed losses.

Minimum Value and Minimizer

Let $g : \mathcal{W} \rightarrow \mathbb{R}$.

Minimum Value

$$\min_{w \in \mathcal{W}} g(w)$$

The smallest objective value.

Minimizer

$$w^* \in \arg \min_{w \in \mathcal{W}} g(w)$$

An allowed input where the minimum occurs.

$$g(w^*) = \min_{w \in \mathcal{W}} g(w).$$

 For a small discrete domain, evaluate every allowed input and compare the objective values.

Discrete Optimization by Comparison

Let $g(w) = w^2$ and $\mathcal{W} = \{-2, -1, 2, 3\}$.

Allowed input w	-2	-1	2	3
Objective value $g(w)$	4	1	4	9

$$w^* = -1, \quad \min_{w \in \mathcal{W}} g(w) = 1.$$

Minimization and Maximization

Negating an objective exchanges minimization and maximization:

$$\arg \min_{w \in \mathcal{W}} g(w) = \arg \max_{w \in \mathcal{W}} [-g(w)].$$

The optimizer stays at the same input, while the optimal value changes sign:

$$\min_{w \in \mathcal{W}} g(w) = - \max_{w \in \mathcal{W}} [-g(w)].$$

Minimizing a Convex Objective

For a differentiable convex function on $\mathbb{R}$,

$$g'(w^*) = 0 \implies w^* \text{ is a global minimizer.}$$

If g is twice differentiable, a useful one-dimensional test is

$$g''(w) \geq 0 \quad \text{for all } w.$$



On a restricted interval, also check the endpoints. A convex function may have more than one minimizer.

Optimization in Several Dimensions

Consider the convex polynomial from the Sep. 22 coupled example:

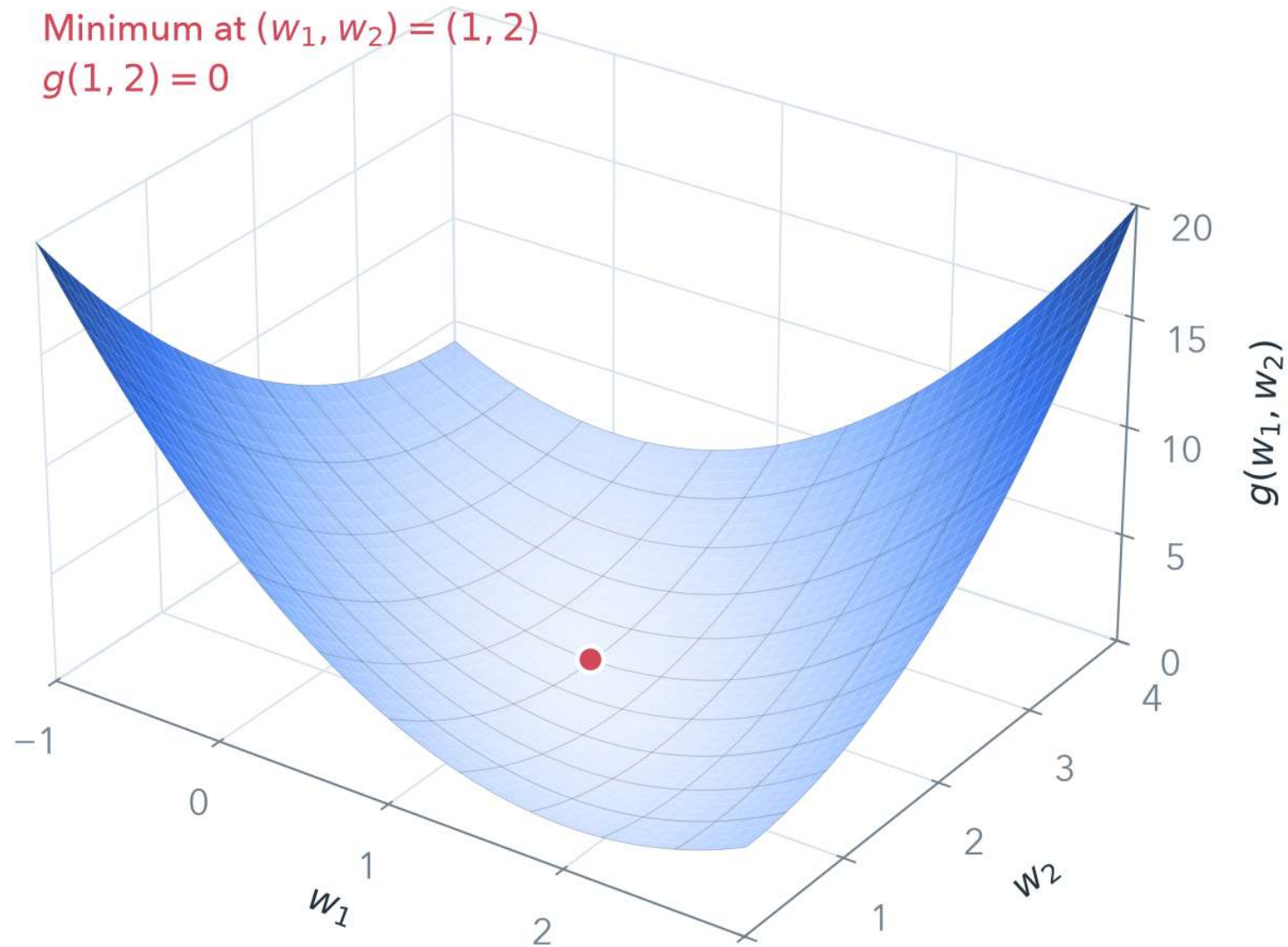
$$g(w_1, w_2) = 2w_1^2 + 2w_1w_2 + w_2^2 - 8w_1 - 6w_2 + 10.$$

Each partial derivative depends on both variables:

$$\frac{\partial g}{\partial w_1} = 4w_1 + 2w_2 - 8, \quad \frac{\partial g}{\partial w_2} = 2w_1 + 2w_2 - 6.$$

- i A minimizer must make both partial derivatives zero at the same point. Solve the two equations together.

Surface of the Coupled Objective



Solving the Coupled Equations

Set both partial derivatives to zero, then divide by 2:

$$\begin{aligned}2w_1 + w_2 &= 4, \\w_1 + w_2 &= 3.\end{aligned}$$

Subtract the second equation from the first:

$$w_1 = 1, \quad w_2 = 3 - w_1 = 2.$$

$$\mathbf{w}^* = (1, 2)^\top, \quad g(\mathbf{w}^*) = 0.$$

i Because g is convex, this stationary point is a global minimizer.

The Closed-Form Linear Regression Learner

Given $\mathcal{D} = ((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n))$,

$$A = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top, \quad \mathbf{b} = \sum_{i=1}^n \mathbf{x}_i y_i, \quad \hat{\mathbf{w}} = A^{-1} \mathbf{b}.$$

The learner returns

$$\hat{f}(\mathbf{x}) = \mathbf{x}^\top \hat{\mathbf{w}}.$$

Review course note and assignments.